

# ZOIE TELKAMP

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## Education

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### University of Virginia

*PhD in Astronomy*

*Master of Science in Astronomy, GPA: 4.0/4.0*

Advisor: Dr. Adele Plunkett

Anticipated Defense Date: April 2027  
2023

### University of California, Berkeley

*Bachelor of Arts in Astrophysics*

2020

## Research Experience

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### Graduate Student Researcher

2021–Present

*University of Virginia, Department of Astronomy*

Investigating star formation from low to high masses using observations from ALMA, JWST, Herschel, Spitzer, and SOFIA. Co-developed open-source analysis tools including the *sedcreator* Python package to enable community science (used in 15+ publications).

### Junior Astronomy Specialist

2020–2021

*University of California, Berkeley*

Led development of an autoencoder neural network to generate synthetic edge-on protoplanetary disk images 1000× faster than traditional radiative transfer calculations, to facilitate efficient disk modeling. Published first-authored paper in *The Astrophysical Journal*.

### Undergraduate Astronomy Researcher

2019–2020

*University of California, Berkeley*

Used methods including principal component analysis to investigate relationships between protoplanetary disk physical properties and morphology. Co-authored two publications and participated in observing runs at Keck and Lick observatories.

## Teaching Experience

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### Graduate Student Teaching Assistant

*University of Virginia, Department of Astronomy*

Astronomy 4470/5470: Computational Astrophysics (Graduate-level)

Spring 2026

Astronomy 1220: Stars, Galaxies, and the Universe

Spring 2026

Astronomy 1210: Intro Sky and Solar System

Spring 2024

Astronomy 1280: Origins of Almost Everything

Spring 2024

Astronomy 5110: Astronomical Techniques (Graduate-level)

Fall 2023

Astronomy 1220: Stars, Galaxies, and the Universe

Fall 2023

Astronomy 2110: Introduction to Astrophysics

Fall 2022

Astronomy 1290: Black Holes

Fall 2022

**Summer Student Supervisor**

Summer 2022

*Chalmers/Virginia Initiative on Cosmic Origins (CICO/VICO)*

Co-supervisor to Sam Crowe, “Massive Protostars in the Galactic Center” and Alejandro Romar Tejeiro, “Infrared Light in the Dark (I-LID)—A Census of Star Formation in IRDCs”.

**Undergraduate Course Grader**

Fall 2019

*University of California, Berkeley*

Astronomy 160: Stellar Astrophysics

## Awards and Honors

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**Jefferson Scholars Foundation Dissertation Year Fellowship**

2026-2027

*Awarded by the Jefferson Scholars Foundation.***Emma Williams Prize in Astronomy**

2025

*Awarded by the University of Virginia Astronomy Department for outstanding research achievement by a fourth-year graduate student.*

**American Astronomical Society (AAS) Nova, “Selections from 2025”**

2025

*Telkamp et al. (2025) featured on Nova as one of the most-read journal articles published that year.*  
<https://aasnova.org/2025/12/22/selections-from-2025-the-formation-of-massive-stars/>

**Virginia Space Grant Consortium (VSGC) Graduate STEM Research Fellowship Program**

2023–2025

*Awarded by the VSGC to help support a research project supporting NASA’s mission.***International Astronomical Union Grant**

2023

*Awarded by the IAU to attend Kavli-IAU S383.***Dorothea Klumpke Roberts Prize in Astronomy**

2020

*Awarded by the UC Berkeley Astronomy Department for outstanding scholarly achievement by either an upper-level undergraduate student, or an outstanding graduate student early in his/her research career.*

**Undergraduate Research Apprentice Program (URAP) Summer Award**

2020

*Awarded to select students nominated by their project mentor to support their summer research.*

## Publications

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*First Author*

**Telkamp, Z.**, Plunkett, A., Tan, J.C., 2026, “An Archival ALMA Survey of Disks around Massive Protostars,” submitted to ApJ; revisions in progress

**Telkamp, Z.**, Fedriani, R., Tan, J. C., Law, C., Yang, Y., Plunkett, A, et al., 2025, “The SOFIA Massive (SOMA) Star Formation Survey. V. Clustered Protostars,” ApJ, 986, 15

**Telkamp, Z.**, Martínez-Palomera, J., Duchêne, G., Ashimbekova, A., Wolfe, E., Angelo, I., Pinte, C., 2022, “A machine learning framework to predict images of edge-on protoplanetary disks,” ApJ,

*Contributing Author*

Fedriani, R., Tan, J. C., **Telkamp, Z.** (+13 co-authors), 2023, “The SOFIA Massive (SOMA) Star Formation Survey. IV. Isolated Protostars,” *ApJ*, 942, 53

Jørgensen, J., Coutens, A., Drozdovskaya, M., Lee, J.-E., Plunkett, A., Maret, S., Belloche, A., Harsono, D., **Telkamp, Z.** (+20 coauthors), 2026, “Complex Organic Molecules in Protostars with ALMA Spectral Surveys (COMPASS). I. Overview of the ALMA Large Program,” *A&A*, in press

Plunkett, A. (+26 coauthors and **Telkamp, Z.**), 2026, “Complex Organic Molecules in Protostars with ALMA Spectral Surveys (COMPASS). II. Approach to data reduction and products for line-rich broadband (sub-)millimeter spectra,” *A&A*, in press

Coutens, A. (+26 coauthors and **Telkamp, Z.**), 2026, “Complex Organic Molecules in Protostars with ALMA Spectral Surveys (COMPASS). III. CH<sub>3</sub>OH isotopic fractionation in the low-mass protostar BHR71-IRS1,” submitted to *A&A*

Gigli, D., Gorai, P., Law, C. Y., Tan, J. C., Bonfand, M., Rahman, T., Zhang, Y., Taniguchi, K., Fedriani, R., **Telkamp, Z.** (+2 co-authors), 2025, “The SOMA MM Survey. I. An Astrochemical Census of Massive Protostars,” *A&A*, in press

Oakey, P., Yang, Y. L., Tan, J. C., Bisbas, T., Fedriani, R., Tanaka, K., **Telkamp, Z.** (+3 co-authors), 2025, “The SOMA Atomic Outflow Survey. I. An Atomic OI and Highly Ionized OIII Outflow from Massive Protostar G11.94-00.62,” *A&A*, in press

Crowe, S., Fedriani, R., Tan, J. C., Whittle, M., Zhang, Y., Caratti o Garatti, A., Farias, J. P., Gautam, A., **Telkamp, Z.** (+24 co-authors), 2024, “Near-infrared observations of outflows and young stellar objects in the massive star-forming region AFGL 5180,” *A&A*, 682, 27

Finn, M., Mazzei, R., Drechsler, B., **Telkamp, Z.**, Rao, M., Agrawal, P., McAlister, A., 2023, “How Informal Science Education Influences Elementary Students’ Perceptions of Science and Themselves,” *Journal of STEM Outreach*, 6, 1

Angelo, I., Duchêne, G., Stapelfeldt, K., **Telkamp, Z.** (+8 coauthors), 2023, “Demographics of Protoplanetary Disks: A Simulated Population of Edge-on Systems,” *ApJ*, 945, 130

Villenave, M., Ménard, F., Dent, W., Duchene, G., Stapelfeldt, K., Benisty, M., Boehler, Y., Plas, G., Pinte, C., **Telkamp, Z.** (+8 co-authors), 2020, “Observations of edge-on protoplanetary disks with ALMA. I. Results from continuum data,” *A&A*. 642, 164

**Talks**


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| “Benchmarking LLM Use in Multiwavelength Astronomy”<br><i>NSF-Simons AI Institute for Cosmic Origins Webinar</i>                                                          | 2026 |
| “An Archival ALMA Survey of Disks Around Massive Protostars”<br><i>Chalmers/Virginia Initiative on Cosmic Origins (CICO/VICO) Spring Workshop, University of Virginia</i> | 2026 |
| “A Multi-Scale Comparison of Star Formation from Low to High Masses”<br><i>Astronomy Seminar, Virginia Tech (Invited)</i>                                                 | 2026 |

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| “Investigating Massive Star Formation with the Atacama Large Millimeter/<br>submillimeter Array (ALMA)”<br><i>Virginia Space Grant Consortium Graduate Research Conference, Hampton University</i>                                      | 2025 |
| “The SOFIA Massive (SOMA) Star Formation Survey: Clustered Regions”<br><i>Bob Rood Symposium, University of Virginia</i>                                                                                                                | 2025 |
| “Probing Massive Star Formation with the ALMA Archive”<br><i>Bob Rood Symposium, University of Virginia</i>                                                                                                                             | 2024 |
| “Zooming into Massive Star Formation: A Multi-scale Approach”<br><i>“TUNA” Lunch Talks, National Radio Astronomy Observatory (NRAO)</i>                                                                                                 | 2023 |
| “The SOMA Survey - Probing Massive Star Formation Across Galactic Environments”<br><i>Our Galactic Ecosystem: Opportunities and Diagnostics in the Infrared and Beyond, University of California, Los Angeles, Lake Arrowhead Lodge</i> | 2022 |
| “The SOFIA Massive (SOMA) Star Formation Survey: New Algorithms and New<br>Environments”<br><i>Virginia Initiative on Cosmic Origins, University of Virginia</i>                                                                        | 2021 |
| “Developing a Physics-Aware Neural Network to Generate Edge-On Protoplanetary<br>Disk Images”<br><i>237th Meeting of the American Astronomical Society, Virtual format</i>                                                              | 2021 |

## Posters

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| “Benchmarking LLM Use in Multiwavelength Astronomy”<br><i>2nd Annual NSF-Simons AI Institute for Cosmic Origins Science Conference, Charlottesville, Virginia</i> | 2026 |
| “Probing Massive Star Formation with the ALMA Archive”<br><i>ALMA at 10 years: Past, Present, and Future, Virtual format</i>                                      | 2023 |
| “The SOMA Survey - Probing Massive Star Formation Across Galactic Environments”<br><i>From Stars to Galaxies II, Chalmers University of Technology</i>            | 2022 |
| “Characterizing the Radial and Vertical Structure of Protoplanetary Disks”<br><i>Summer Intern Poster Symposium, University of California, Berkeley</i>           | 2019 |

## Leadership and Outreach

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| <b>Sedcreator Python package</b><br><i>Co-lead developer</i>                                                                                                                                                           | 2021–Present |
| <b>Graduate School of Arts and Sciences Council</b><br><i>Representative for the UVA Astronomy Department</i>                                                                                                          | 2023–2024    |
| <b>Dark Skies, Bright Kids (DSBK) Assessments</b><br><i>Researcher and co-author of a publication in the Journal of Stem Outreach investigating the enrichment of science education in under-served school systems</i> | 2021–2023    |

## Additional Skills

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**Technical:** Python, Docker, UNIX, HTML, JavaScript, LaTeX, GitHub, Google Colab, SAOImageDS9, MCFOST

**AI/Machine Learning:** Large Language Models (LLMs), Deep Neural Networks, Principal Component Analysis (PCA), Random Forest Regression, Multivariate Linear Regression, Polynomial Regression